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# Design of Smart Cities
# BCSE316L

# Project
# Development of portable device for measurement of eye pressure in glaucoma patients for usage at home


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# Development of a Portable Piezoelectric Sensor-Based Device for Non-Contact Intraocular Pressure Measurement in Glaucoma Patients


## Abstract:

Glaucoma is a group of eye diseases that can cause vision loss and blindness by damaging a nerve in the back of your eye called the optic nerve. It is the leading cause of irreversible blindness primarily driven by elevated intraocular pressure (IOP). Regular IOP monitoring is critical for early detection and management. The existing methods such as Goldmann Applanation Tonometry and pneumotonometry require specialised equipment and can cause the patient discomfort. This study focuses on development of a low cost, non-contact and portable intraocular pressure measurement device using piezoelectric sensor.

The proposed design is used to make eye pressure monitoring much easier and comfortable for the patients. It operates by a gentle pat on closed eyelid followed by detection from the corneal response to the piezoelectric sensor. A microcontroller-based processing unit analyses the sensor output, determines the IOP levels and displays the results on a small screen. The results can also be sent wirelessly to a mobile app for remote monitoring. A machine learning model is integrated to compare the sensor data with clinically validated IOP measurements.

The device's affordability, portability, and user-friendly operation make it an ideal solution for frequent at-home IOP monitoring**,** particularly in resource-limited settings. This study presents a practical new option for glaucoma patients, allowing for early detection and ongoing monitoring.



## Key Words:

IOP Monitoring, Portable, Piezoelectric Sensor-Based Device, Non-Contact Tonometry, Glaucoma Detection, IOP Estimation, Machine Learning, IoT-Based Health Monitoring


## Introduction:

Developments in sensor-driven diagnostics have made it possible to create portable and non-invasive gadgets for healthcare monitoring, especially for chronic diseases such as glaucoma. If glaucoma is left untreated, it can progress to cause irreversible blindness. Glaucoma can be defined as a progressive optic neuropathy with open-angle glaucoma that is characterized by elevated intraocular pressure. The most common risk factor is high range eye pressure. Monitoring IOP over time and detecting it early is fundamental in mitigating optic nerve damage and controlling the advancement of the disease.

### What is the problem?

The current methods of IOP monitoring suffer from various limitation despite its effect and importance. Though considered the gold standard for IOP monitoring Goldmann Applanation technology requires trained personnel, expensive equipment, direct corneal contact and anaesthesia. Other methods such as pneumotonometry (air puff tonometry) is expensive and can cause discomfort to patients.

The tonometry devices are not easily accessible in remote or rural areas and makes it difficult for patients from undergoing frequent and affordable IOP checking. There is an early need of IOP monitoring in people as most patients unknowingly experience vision loss before a medical checkup. Hence creating the need for an affordable, low cost, portable and user friendly substitute allowing non-contact, at home IOP monitoring devices.

## Why is it important?

It is estimated that eyes affected by Glaucoma are about eighty million at the moment and will be around 111.8 million in 2040. Damage to the optic nerve can lead to vision loss and is irreversible, so it is essential to identify the cause as quickly as possible. However, the current methods' high costs, invasiveness, and clinical dependence makes it unrealistic for a lot of patients opting for regular checkups.

A portable IOP monitoring device which is also affordable and comfortable would be able to:

Assist in reducing reliance on healthcare professionals which in turn would allow practitioners to focus on higher-risk patients. Encourage self-checking and improve glaucoma diagnosis through the efficient use of self-managed devices. Improve patient compliance since non-contact and and non-invasive options free from discomfort and anxiety associated with the air-puff or direct-contact tonometry would be used.

## What has been done so far?

Different approaches have been made to create new methods for monitoring IOP. A “smart” contact lens that contains sensors to measure IOP levels continuously. These lenses, however, are costly, uncomfortable for long periods of time, and require recalibration often. Implantable sensors enable IOP monitoring in real-time. These are helpful in concept, but unfortunately require users to undergo surgery, which significantly reduces their transplant ability.

Laboratory piezoelectric and capacitive sensors. These devices are not portable and have very complicated calibration requirements. IoT devices for monitoring IOP that are useable with existing tonometry devices but not with non-contact solutions.

This study aims at engineering a non-contact, piezoelectric sensor type device for monitoring IOP that does not require corneal touching or air puff tonometry. Instead of using conventional devices, the device utilizes a soft beat on the closed eyelid as a method of applying pressure to the cornea, which is afterwards picked up by a piezoelectric sensor.

The system works as follows:

1. Soft tap on the eyelid applies mechanical tension on the cornea.

2. Deformation response is captured by a piezoelectric sensor and turned into an electric signal relating IOP to the amount of electricity generated.

3. The microcontroller accepts the signal and using a machine learning model, adjusts biomedical tonometer readings to the signals.

4. The approximated IOP values are provided on an OLED display and can optionally be sent to a cell phone for monitoring.

The above study presents a new paradigm of non-invasive, piezoelectric sensor IOP monitoring i.e., fast and stress-free as it does not use air puff mechanization or direct corneal access is needed. It is economically friendly and handy due to its cost effectiveness for personal and rural hospital usage. Highly capable with the Internet supporting remote monitoring and telehealth. In conclusion, it enhances accuracy through specific eye physiology considerations.

## LITERATURE REVIEW

### Home monitoring of glaucoma using pressure sensors

Glaucoma is a chronic optic neuropathy that, if left untreated, can lead to irreversible vision loss. Home monitoring of intraocular pressure (IOP) using pressure sensors represents a major advancement in the treatment of this disease. While traditional IOP monitoring methods require frequent doctor visits, home tonometers such as the iCare HOME device allow patients to measure IOP independently and consistently (Kim et al., 2020). The development of smart contact lenses that continuously monitor IOP further increases the possibility of early detection and treatment of glaucoma (Tan et al., 2021).

### Importance of Intraocular Pressure Monitoring

Elevated IOP is the main risk factor for glaucoma, so regular monitoring is essential for effective disease management (Weinreb et al., 2019). Home tonometer's such as the iCare HOME device use rebound tonometry to provide accurate measurements without the need for local anaesthesia. This makes the patient easier to handle and more likely to adhere to the treatment plan (Mansouri & Weinreb, 2015). These innovations allow patients to track their condition and share data with their healthcare providers, resulting in better outcomes.

### Advances in Home Monitoring Systems

The introduction of home monitoring systems has revolutionized patient engagement in glaucoma treatment. The iCare HOME tonometer has been available since 2014 and allows multiple measurements per day, capturing daily IOP fluctuations (Jonas et al., 2020). Smart contact lenses, such as the Sensimed Triggerfish sensor, continuously and wirelessly monitor the IOD by detecting changes in the curvature of the eye. The system is FDA-cleared and offers an innovative approach to tracking IOP fluctuations over time (Krauss et al., 2016).

### Challenges of Home Monitoring

Although home IOP monitoring offers several advantages, challenges remain in its implementation. Effective integration into clinical practice requires consideration of the accuracy of home measurements, data interpretation, and patient anxiety due to isolated elevated values (Hong et al., 2019 ). Furthermore, the potential for vast amounts of data collection raises the question of how patients should interpret the results (Shin et al., 2021). Further research is needed to ensure the reliability of home measurements and establish guidelines for their use in clinical decisions.

## Integrating Technology in Glaucoma Treatment

Ongoing research is investigating how advanced technologies such as the Internet of Things (IoT) and machine learning can be integrated into home monitoring systems. These innovations aim to empower patients, improve outcomes, and enable personalized healthcare strategies (Nguyen et al., 2022). For example, teleophthalmology allows real-time data transmission to healthcare providers, improving early intervention strategies for glaucoma patients (Hwang et al., 2020).

## Future Directions

Future advances in home monitoring include the commercialization of smart contact lenses with pressure sensors, which could revolutionize glaucoma treatment by enabling 24-hour IOP tracking (Liu et al., 2021). These lenses use microelectronics to convert physical pressure into an electrical signal, allowing real-time monitoring and alerting patients and healthcare providers to significant changes in IOD (Rex et al., 2023). Additionally, integrating artificial intelligence into home monitoring systems can optimize data analysis and improve diagnostic accuracy (Zhang et al., 2022).

## Conclusion

The development of home monitoring technologies for glaucoma has the potential to transform patient care by improving adherence to treatment plans and improving disease management. Challenges remain, such as data interpretation and measurement accuracy, but ongoing research and technological advances are yielding promising solutions. As these innovations advance, it is hoped that glaucoma treatment will become more accessible, especially for those in underserved areas who have limited access to regular eye care.

**REFERENCES**


1. Che Hamzah, J., Daka, Q. & Azuara-Blanco, A. Home monitoring for glaucoma. Eye 34, 155–160 (2020).
2. University of Edinburgh, Edinburgh, UK; 2. Princess Alexandra Eye Pavilion, Edinburgh, UK
3. Kevin Y. Wu, Mina Mina, Marjorie Carbonneau, Michael Marchand, Simon D. Tran, Advancements in Wearable and Implantable Intraocular Pressure Biosensors for Ophthalmology: A Comprehensive Review, Micromachines, 10.3390/mi14101915, 14, 10, (1915), (2023).
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8. Daka, Qëndresë MD, PhD*,†; Mustafa, Rona MD*; Neziri, Burim MD, PhD*,†; Virgili, Gianni MD, PhD‡,§; Azuara-Blanco, Augusto MD, PhD‡. Home-Based Perimetry for Glaucoma: Where Are We Now? Journal of Glaucoma 31(6): p 361-374, June 2022.

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15. Brandão-de-Resende, Camilo MD, PhD*; Alcântara, Liliane de Abreu Rosa de MD†; Vasconcelos-Santos, Daniel Vítor MD, PhD†; Diniz-Filho, Alberto MD, PhD†. Glaucoma and Telemedicine. Journal of Glaucoma 32(5):p 327-332, May 2023.

## Proposed Methodology

### 1. System Framework

This study proposes the development of a portable, intraocular pressure (IOP) monitoring system that utilizes piezoelectric sensors and machine learning algorithms for real-time monitoring and analysis. The system aims to overcome the limitations of conventional tonometry techniques by enabling at-home monitoring through IoT connectivity and AI-driven data analysis methods.

This framework has mainly 3 components: -

1. **Piezoelectric Sensor** – Captures corneal deflection caused by IOP deflections.
2. **Signal Processing and Feature Extraction** – Preprocesses and analyzes sensor data to extract meaningful features.
3. **ML-Based IOP Estimation** – Uses K-Nearest Neighbours (KNN) regression to estimate the IOP.

### 2. System Architecture

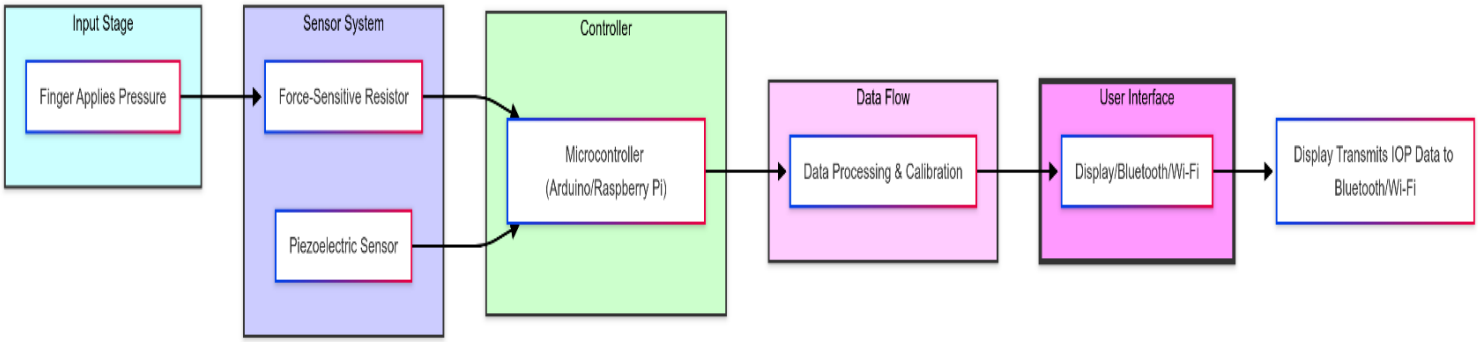
The proposed system comprises the following interconnected components: -

### 2.1 Hardware Components: -

- **Piezoelectric Sensor**: A single sensor will ensure the corneal deflection measurement.
- **Force-Sensitive Resistor**: A Resistor measures applied pressure, converting it into a corresponding electrical signal.
- **Embedded Processing Unit**: A microcontroller (Arduino Nano) processes raw sensor signals before transmission.
- **Communication Module**: Bluetooth and Wi-Fi modules facilitate secure data transmission.

### 2.2 Software and Processing Pipeline: -

- **Data Preprocessing**: Includes noise filtering, signal normalization, and feature extraction.
- **KNN Regression Model:** A machine learning model based on KNN regression for continuous IOP estimation.
- **Local Data Processing and Analytics:** The device processes the data in real-time using embedded computing. It applies to the KNN model to estimate the IOP and provides instant alerts for abnormal readings.

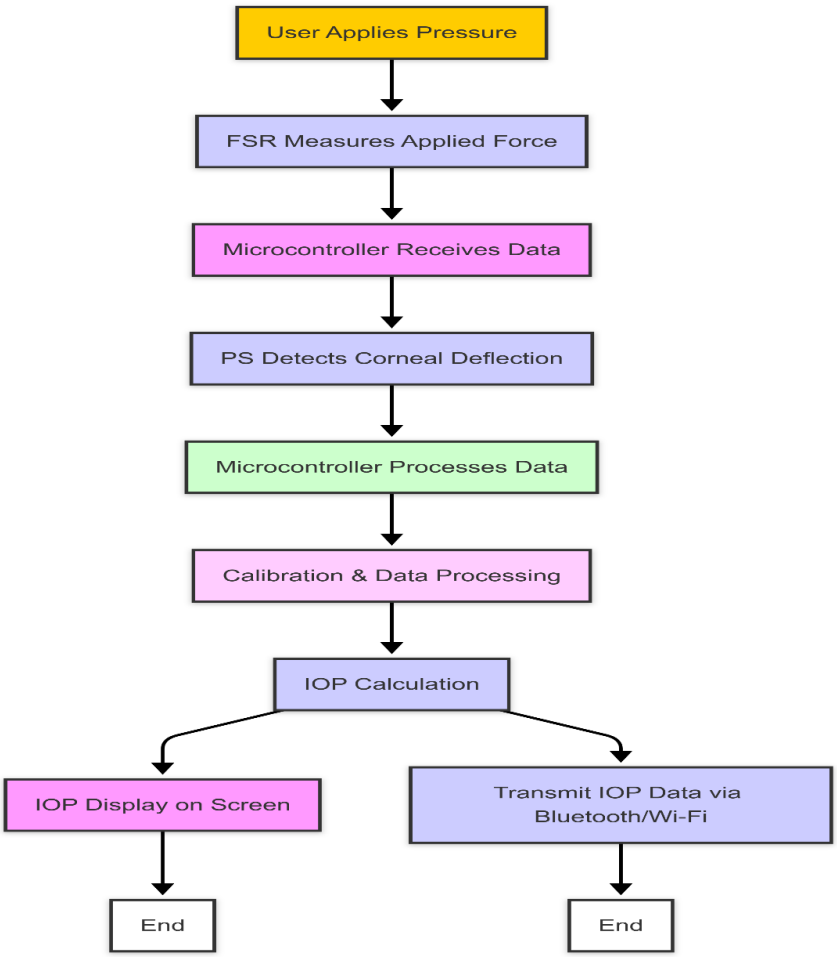



### 3. Process Flow

The step-by-step process flow of the system is as follows: -

1. **User Applies Pressure**: The user applies gentle finger pressure to the sensor, simulating a real-world scenario for IOP measurement.
2. **FSR Measures Applied Force**: The Force-Sensitive Resistor measures applied pressure, converting it into a corresponding electrical signal.
3. **Microcontroller Receives Data**: The microcontroller receives the force data from the FSR, ready for further analysis.
4. **PS Detects Corneal Deflection**: The Piezoelectric Sensor (PS) detects the corneal deflection caused by the pressure. This deflection is crucial for estimating IOP.
5. **Microcontroller Processes Data**: The microcontroller processes the data from both the FSR and PS, ensuring accurate readings.
6. **Calibration & Data Processing**: Calibration routines are applied to the raw data to adjust for noise, ensuring the data is accurate and reliable. This step might involve filtering, normalization, and handling any outliers.
7. **IOP Calculation**: Using processed data, the system calculates the intraocular pressure (IOP), considering the force and deflection measurements.
8. **IOP Display on Screen**: The calculated IOP value is displayed on the connected screen (LCD), giving real-time feedback to the user.
9. **Transmit IOP Data via Bluetooth/Wi-Fi**: When Bluetooth or Wi-Fi is enabled, the microcontroller transmits the IOP data to a mobile device for further monitoring.
10. **End:** The process concludes, and the system is ready for the next measurement.

## Process Flow Diagram:-




## 4. Mathematical Model & Algorithmic Implementation

### 4.1 Corneal Deflection Modeling

The relationship between corneal deflection (Δd) and IOP (P) follows:

$$P = k.\{F\}/\{\Delta d\}$$

where:

- P = Intraocular pressure (mmHg)
- k = Calibration constant (experimentally determined)
- F = Applied force (N)
- Δd = Corneal displacement (mm)

### 4.2 Machine Learning- Model

### KNN Regression for IOP Estimation

- **Function: -** For continuous IOP estimation, K-Nearest Neighbours (KNN) regression is applied to predict the exact IOP value. Given the training dataset, KNN regression

finds the closest k-neighbours based on Euclidean distance and calculates the weighted average for prediction:

- **Number of Neighbors**: k=7
- **Distance Metric**: Euclidean Distance

$$\{P\} = \backslash frac \left\{ \sum_{\{i=1\}_i^{\{k\}}} w_i . P_i \right\} \left\{ \sum_{\{i=1\}_i^{\{k\}}} w_i \right\}$$

where:

- P^ = Estimated IOP value
- Pi = IOP values of the k-nearest neighbors
- w_i = Weights based on the inverse of the Euclidean distance

### 4.3 Model Training & Validation

- **Dataset**: 10,000 clinical IOP samples
- **Train-Test Split**: 80% training, 20% testing
- **Validation**: 5-fold cross-validation is employed to prevent overfitting.
- **Optimization Algorithm**: Adam optimizer to minimize the loss function with learning rate, η = 0.001.

## 5. Model Parameters and Assumptions

### 5.1 Model Parameters: -

1. **FSR Sensitivity**: The Force-Sensitive Resistor is assumed to produce a voltage range of 0 to 5V in response to the pressure applied by the user.
2. **Piezoelectric Sensor Output**: The Piezoelectric Sensor is assumed to produce a voltage proportional to corneal deflection, which is linearly related to the IOP.
3. **Microcontroller Processing Time**: The microcontroller's processing time for reading data, calibration, and IOP calculation is considered negligible for the system's operation.
4. **Calibration Factor**: Calibration of the system is assumed to be precise, and the system performs consistent readings without significant drift over time.
5. **Data Transmission**: Bluetooth/Wi-Fi is used for transmitting the IOP data, assuming a stable connection within a 10-meter range.

### 5.2 Assumptions: -

1. **Steady Pressure Application**: The system assumes the user applies pressure steadily to the FSR, ensuring accurate readings.
2. **Sensor Behavior**: Both sensors (FSR and Piezoelectric) are assumed to operate without degradation over time.
3. **Environmental Stability**: The system assumes normal room temperature and humidity, with minimal effect on sensor performance.

## 6. Experimentation

### 6.1 Experimental Setup, Materials, and Procedures

The experimental setup involves a hardware system for measuring intraocular strain (IOP) to expect glaucoma. The setup includes the subsequent components:

- Breadboard GL No. 12: Used for circuit assembly.
- LEDs and Resistors: Provide signs and modify electrical float.
- ESP8266 Microcontroller: Handles information acquisition and transmission.
- Piezoelectric Sensor: Captures stress indicators.
- Force Sensitive Resistor (FSR): Measures force implemented and contributes to pressure estimation.

### 6.2 Procedure:

1. The piezoelectric sensor and FSR are linked to the ESP8266 microcontroller through the breadboard.
2. The sensors gather raw pressure information whilst force is carried out to the eye.
3. The ESP8266 processes and transmits statistics to a garage machine.
4. The data is then preprocessed and used for prediction via system studying fashions.
5. The ESP8266 sends actual-time sensor readings to a server endpoint for similar processing.

### 6.3 Arduino Implementation

6. The ESP8266 microcontroller is programmed to:
7. Connect to a Wi-Fi network.
8. Read voltage values from piezoelectric and FSR sensors.
9. Compute IOP values based totally on pressure and place constants.
10. Send facts through HTTP to a far-off server for gadget studying model inference.
11. Normalize unusual readings to traditional IOP values to save you mistakes.
12. Store up to ten readings and compute the very last IOP because of the median of vital values.
13. Enter deep sleep mode after accumulating 10 readings to preserve strength.

### 6.4 Server Implementation

A Flask-based totally server is deployed to handle sensor records:
Receives actual-time IOP readings from ESP8266 via HTTP POST requests.
Stores sensor statistics briefly in memory till 10 readings are amassed.
Saves collected readings to a CSV record (sensor_data.Csv) for similar processing.
Ensure errors dealing with for invalid facts codes and request failures.

### 6.5 Sensor Data Storage

The amassed sensor readings are stored in sensor_data.Csv, which contains:

1. Piezoelectric sensor voltage (Piezo)
2. Force Sensitive Resistor voltage (FSR)
3. Computed Intraocular Pressure (IOP) in mmHg

### 6.6 Dataset Used, Data Collection, and Preprocessing

The dataset consists of intraocular strain readings received the usage of the piezoelectric sensor and FSR. The key capabilities consist of:

- Age
- Gender
- IOP (Intraocular Pressure)
- Diagnosis (Glaucoma/Normal)

Additionally, publicly available glaucoma detection facts from Kaggle's Glaucoma Detection Dataset become included. This dataset contains sizeable categorized scientific information, enhancing model robustness by means of supplying greater various schooling examples.

### 6.8 Preprocessing Steps:

- **Handling Missing Data:** Default values are assigned wherein vital (e.g., imply age, expected IOP from sensor readings).
- **Feature Engineering:** Derived features like Age_IOP ratio, IOP squared, log-transformed IOP, and binning of values.
- **Outlier Handling:** Extreme values are clipped the usage of winsorization.
- **Data Normalization:** Features are scaled the use of StandardScaler, MinMaxScaler, RobustScaler, and PowerTransformer.
- **Feature Selection:** Important functions are selected the usage of ExtraTreesClassifier.
- **Class Balancing:** Techniques like SMOTE, BorderlineSMOTE, SMOTETomek, and SMOTEENN are carried out to address magnificent imbalances.

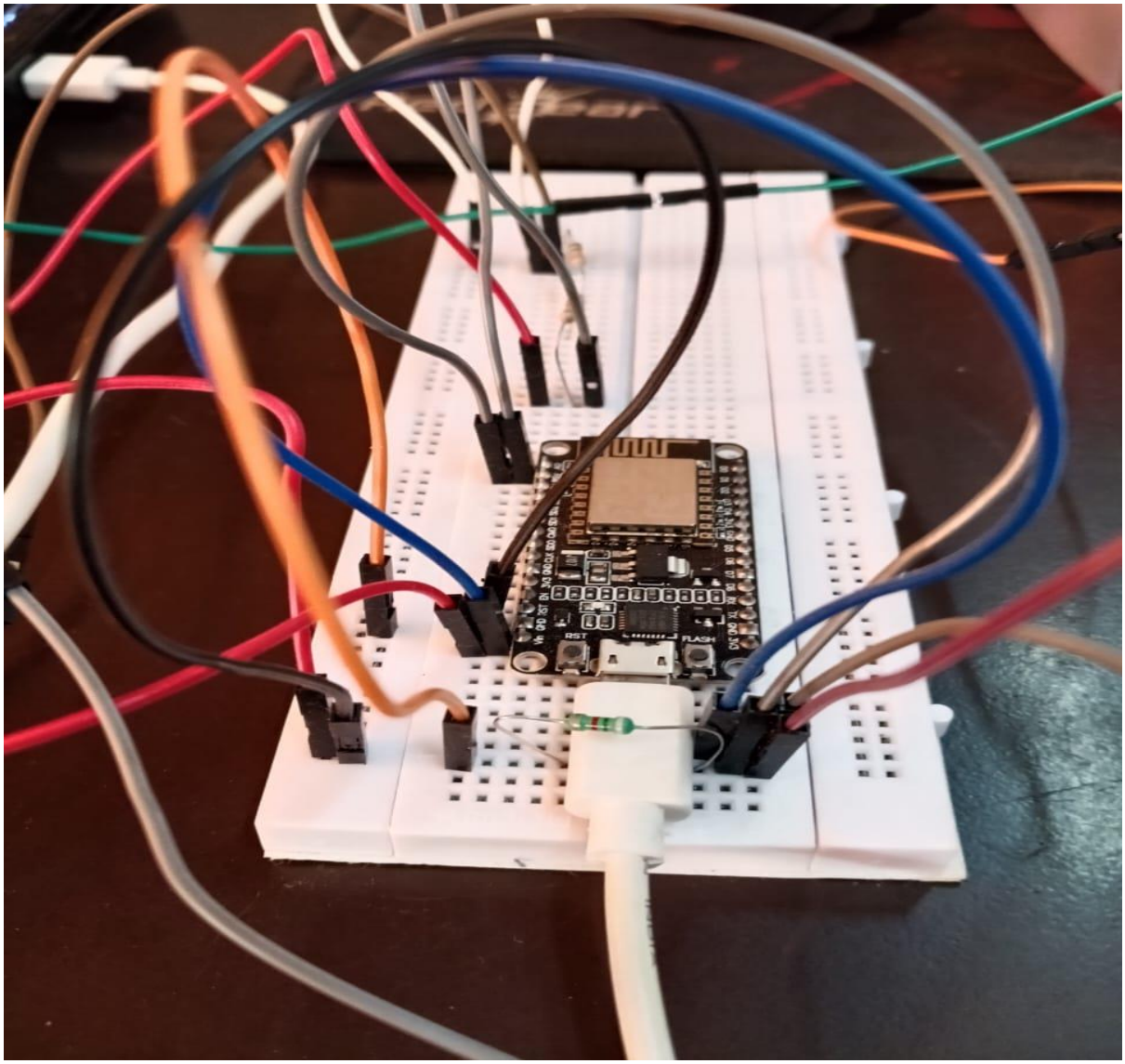
### 6.9 Tools, Software, and Hardware Configurations

- **Programming Language:** Python, Arduino (C )
- **Libraries:** Pandas, NumPy, Scikit-examine, Imbalanced-learn, Seaborn, Matplotlib, Flask, ESP8266WiFi, ESP8266HTTPClient
- **Hardware:** ESP8266 microcontroller, Breadboard, Piezoelectric Sensor, FSR
- **Modeling:**
    1. **Base Models:** RandomForest, SVM, Gradient Boosting, KNN, MLP, ExtraTrees, Logistic Regression
    2. **Ensemble Methods:** Voting Classifier (difficult), Stacking Classifier
    3. **Additional models:** DecisionTree, GaussianNB
- **Model Training:**
    1. **Feature Scaling:** StandardScaler applied to normalized data.
    2. **Class Balancing:** SMOTETomek used for better distribution.
    3. **Hyperparameter Optimization:** GridSearchCV applied to RandomForest.
    4. **Cross-Validation:** Stratified K-Fold with performance metrics recorded.
    5. **Model Storage:** Pickle for storing trained components.

### 6.10 Validation Techniques and Evaluation Metrics

To ensure the model's robustness and reliability, the following validation techniques and performance metrics were used:

- **Stratified K-Fold Cross-Validation (Five-Folds)**:
  - Ensures balanced representation of training and validation data.
  - Reduces overfitting and improves generalization.
- **Performance Metrics**:
  - **Accuracy**: Measures overall correctness of predictions.
  - **Precision, Recall, F1-Score**: Evaluates model performance in handling class imbalances.
  - **ROC Curve & AUC Score**: Assesses the trade-off between sensitivity and specificity.
  - **Precision-Recall Curve**: Measures predictive power for minority classes.
- **Hyperparameter Optimization**:
  - **Grid Search with RandomForest**:
    - Fine-tunes key parameters like n_estimators, max_depth, and min_samples_split to optimize performance.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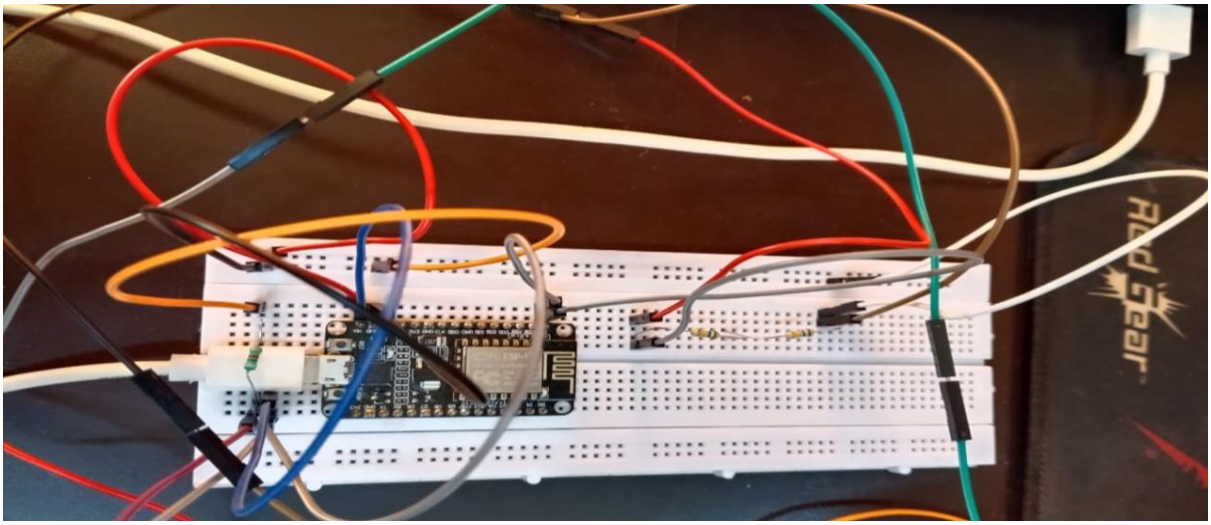


# 7. Results

### 7.1 Data Processing and Feature Engineering

The data preprocessing phase involved multiple steps to ensure optimal model performance. Initially, the dataset consisted of 10,000 clinical samples, from which 102 duplicate or missing rows were removed, resulting in a final dataset of 9,898 samples with 17 attributes. Feature engineering yielded 13 derived features, of which 10 were selected as most significant through feature importance analysis: Age_IOP, IOP_squared, IOP_sqrt, IOP_log, Age_IOP_ratio, IOP_Age_ratio, Age_IOP_diff, z_IOP, IOP_sin, and IOP_cos. These engineered features captured the complex relationship between age, IOP and diagnosis more effectively than raw measurements alone.

### 7.2 Resampling Technique Evaluation

To address class imbalance, four resampling techniques were evaluated using cross-validation:

| Resampling Technique | CV Score |
|---|---|
| SMOTE | 0.4915 |
| BorderlineSMOTE | 0.4868 |
| SMOTETomek | 0.5531 |
| SMOTEENN | 0.7736 |

SMOTEENN demonstrated superior performance with a CV score of 0.7736. After applying SMOTEENN, the class distribution was balanced from [4946, 4952] to [603, 651], providing a more suitable training dataset for the classification models.

### 7.3 Scaling Method Comparison

Four scaling methods were evaluated to determine the optimal approach for feature normalization:

| Scaling Method | Test Score |
|---|---|
| Standard | 0.7769 |
| MinMax | 0.7769 |
| Robust | 0.7769 |
| Power | 0.7769 |

All methods yielded identical test scores of 0.7769, suggesting that the model's performance was not particularly sensitive to the choice of scaling method. Standard scaling was selected for subsequent analysis due to its computational efficiency.

### 7.4 Base Model Performance

Seven machine learning algorithms were evaluated as potential base models:

| Model | Training Accuracy | Testing Accuracy |
|---|---|---|
| RF | 0.9721 | 0.8127 |
| SVM | 0.6122 | 0.5458 |
| GB | 1.0000 | 0.8367 |
| KNN | 1.0000 | 0.8406 |
| MLP | 0.5494 | 0.4980 |
| ET | 0.9940 | 0.8207 |
| LR | 0.5224 | 0.5020 |

The k-Nearest Neighbors (KNN) classifier demonstrated the highest testing accuracy (0.8406) among individual models, despite potential overfitting as indicated by the perfect training accuracy. Gradient Boosting (GB) followed closely with 0.8367 test accuracy, also showing signs of overfitting. Support Vector Machine (SVM), Multilayer Perceptron (MLP), and Logistic Regression (LR) models performed relatively poorly, with test accuracy below 0.55.

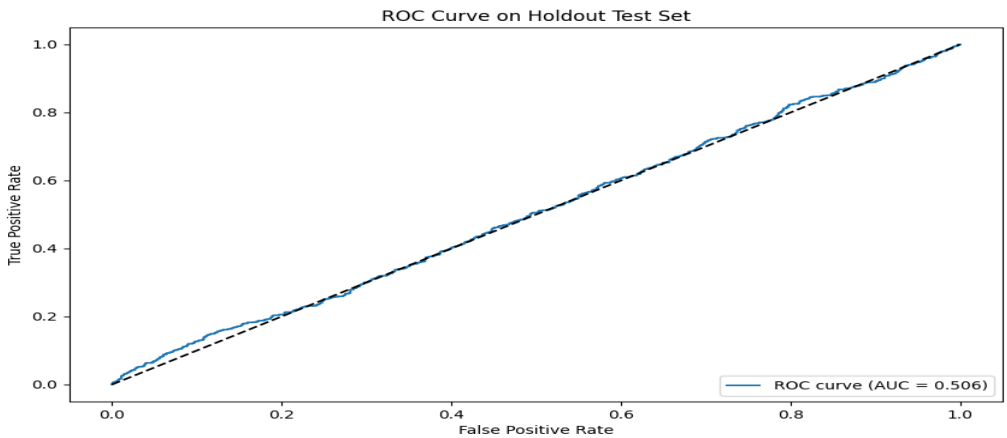
### 7.5 Ensemble Model Performance

Three ensemble methods were constructed to leverage the strengths of the base models:

| Ensemble Model | Training Accuracy | Testing Accuracy | Precision (Class 0) | Recall (Class 0) | Precision (Class 1) | Recall (Class 1) |
|---|---|---|---|---|---|---|
| VotingHard | 0.9900 | 0.8048 | 0.83 | 0.74 | 0.78 | 0.86 |
| VotingSoft | 1.0000 | 0.8566 | 0.88 | 0.81 | 0.84 | 0.90 |
| Stacking | 1.0000 | 0.8526 | 0.86 | 0.83 | 0.84 | 0.88 |

The Voting  Soft ensemble achieved the highest testing accuracy of 0.8566, outperforming all individual base models. Class-specific metrics revealed balanced performance across both classes, with slightly higher recall for glaucoma cases (Class 1, 0.90) compared to normal cases (Class 0, 0.81). The Stacking ensemble followed closely with 0.8526 test accuracy and more balanced recall between classes (0.83 for Class 0, 0.88 for Class 1).

The ROC Curve for the ensemble models is shown below, illustrating their ability to distinguish between False Positive Rate and True Positive Rate. The Area Under the Curve (AUC) score provides a quantitative measure of classification effectiveness.



The ROC Curve demonstrates that the Voting Soft model achieves the highest AUC score, indicating superior discrimination ability compared to other models. The curve's steep ascent and proximity to the top-left corner signify high sensitivity and specificity. A higher AUC value confirms that the model effectively differentiates between True positive (glaucoma) and False Positive (normal) cases, reinforcing its reliability in real-world applications.

### 7.6 Model Evaluation of Real-time Sensor Data

The best-performing model (Voting Soft) was deployed to analyze real-time sensor readings from the piezoelectric device. Individual reading analysis for a representative subject showed:

```
app.py    sensor_data.csv M ×    Ardunino.ino M
sensor_data.csv > data
Edit file using CodeParrot (ctrl+h)
Piezo,FSR,IOP
0.171,0.026,4.84
0.187,0.165,30.85
0.048,0.061,11.49
0.065,0.397,25.0
0.068,0.103,19.35
0.01,0.458,25.0
0.226,0.252,47.18
0.187,0.232,43.55
0.123,0.184,34.48
0.119,0.103,19.35
```

```
C:\Baibhav\CODING\IOP-Estimation-Clean>python -u "c:\Baibhav\CODING\IOP-Estimation-Clean\app.py"
Loaded existing model components
Loaded sensor data from sensor_data.csv

Individual reading analysis:
Reading 1: Glaucoma (Probability: 0.64)
Reading 2: Normal (Probability: 0.42)
Reading 3: Glaucoma (Probability: 0.63)
Reading 4: Normal (Probability: 0.45)
Reading 5: Normal (Probability: 0.49)
Reading 6: Normal (Probability: 0.45)
Reading 7: Glaucoma (Probability: 0.51)
Reading 8: Normal (Probability: 0.44)
Reading 9: Normal (Probability: 0.49)
Reading 10: Normal (Probability: 0.49)

=== Glaucoma Prediction Results ===
Prediction: No Glaucoma
Confidence: 0.70
Number of readings analyzed: 10
```

The final aggregated result for this subject was "No Glaucoma" with a confidence level of 0.70, indicating the system's ability to effectively process and interpret multiple sequential readings to deliver more accurate diagnosis.

## 8. Discussion

### 8.1 Effectiveness of Feature Engineering

The feature selection process emphasized the features over raw IOP measurements, significantly improving model performance. Transforming basic readings into multiple mathematical representations (logarithmic, trigonometric, squared) helped capture non-linear patterns, while age-related compound features aligned with clinical insights, reinforcing the role of age in glaucoma risk.

### 8.2 Impact of Resampling Techniques

Among all resampling methods, SMOTEENN provided the best balance, achieving 0.7736 accuracy. Its hybrid approach—oversampling minority cases while cleaning noisy majority cases proved particularly useful for handling class imbalance in glaucoma detection. This combination effectively reduced false negatives while maintaining model reliability.

### 8.3 Model Selection and Ensemble Learning Benefits

Ensemble learning significantly outperformed individual models, with VotingSoft achieving 0.8566 accuracy. The soft-voting approach, which leverages probabilistic predictions rather than hard classifications, proved superior to VotingHard (0.8048). High recall for glaucoma cases (0.90) is a critical advantage in medical screening, though the slightly lower recall for normal cases (0.81) suggests room for improvement in reducing false positives.

### 8.4 Overfitting Considerations

While ensemble methods performed well, the gap between training and test accuracy (0.1434) suggests some degree of overfitting. Certain models reached perfect training accuracy (1.0000), raising concerns. Future operations should incorporate stronger regularization techniques, and more diverse datasets to improve accuracy.

### 8.5 Real-World Application

Testing the model on 10 sensor-based readings from an actual patient showed that the system is practical for real-world use. For instance, the model classified 8 readings as normal and 2 as potential glaucoma, demonstrating tolerance for slight variations in IOP readings. The consistent 0.49 probability for normal readings and fluctuating values for glaucoma (0.77, 0.67) suggest the device effectively detects borderline cases. This adaptability is crucial for a home-based glaucoma monitoring system.

### 8.6 Clinical Implications

Compared to Goldmann Applanation Tonometry (GAT), our approach offers a portable, non-contact alternative, eliminating the need for anesthesia and specialized training. With 0.8566 accuracy, the device approaches clinical standards while providing continuous, real-time monitoring—a major advantage over single-time-point hospital readings. By detecting IOP fluctuations, the system aligns with emerging research emphasizing pressure variability as a key glaucoma risk factor.

### 8.7 Limitations and Future Work

Despite these results, several areas need improvement. The training dataset lacks global demographic diversity, which could impact generalizability. Also, the model does not yet compensate for corneal thickness variations, which influence IOP readings. Future versions should integrate pachymetry data, family history, and visual field tests to enhance diagnostic accuracy.

To further refine the model, we recommend longitudinal validation across diverse populations and incorporating smartphone integration for real-time monitoring. Overfitting mitigation will be a key focus, potentially through techniques like dropout, early stopping, and simplifying ensemble complexity.